

STUDENT NAME



YEAR 12
Mathematics Extension 2
HSC Course
Assessment Task 3
June 2010

1. There are 2 questions.
2. Marks allocated to each question are indicated in brackets
3. Answer each question on your own paper showing all necessary working
4. Start each question on a new page
5. Calculators may be used
6. Time allowed - **70 minutes**

Topic	Mark
Question 1 (Integration)	/22
Question 2 (Conics)	/22

TOTAL **/44**

Question 1 Integration (marks are shown in brackets)

a) Find $\int \frac{x}{\sqrt{x^2+4}} dx$ (2)

b) Find $\int \frac{\cos^{-1}x}{\sqrt{1-x^2}} dx$ (2)

c) Find $\int \frac{dx}{e^x + e^{-x}}$ (2)

d) Find $\int \frac{1}{\sqrt{x^2+4x-12}} dx$ (3)

e) Find $\int \frac{1}{5+3\cos x} dx$ (3)

f) Let $I_n = \int_0^2 x^n e^{-x} dx$, where n is a **non-negative** integer.

i. Show that $I_n = -2^n e^{-2} + n I_{n-1}$ (3)

ii. Evaluate I_2 (2)

g) Sketch the graph of the curve $y = \frac{1}{x(3-x)}$ for $0 < x < 3$.

Find the area bounded by the curve, the X axis and the lines $x = 1$ and $x = 2$. (5)

Question 2 Conics

- a) The equation of a conic is $\frac{x^2}{4} + \frac{y^2}{3} = 1$ Find
- The eccentricity
 - The coordinates of the foci
 - The equations of the directrices.
- (3)
- b) Find the equation of the chord of contact of the tangents to the hyperbola $x^2 - 16y^2 = 16$ from the point with coordinates $(2, -4)$.
- (2)
- c) Show the area of the ellipse $x = a \cos \theta, y = b \sin \theta$ is πab
- (4)
- d) The points $P(cp, \frac{c}{p})$ and $Q(-cp, \frac{-c}{p})$ lie on the hyperbola with equation $xy = c^2$. The normal at P meets the hyperbola again at R. Show that $\angle PQR$ is a right angle.
- (4)
- e) The point $P(x_1, y_1)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The tangent at P intersects the X axis at A and Y axis at B.
- Find the coordinates of A and B
- (2)
- Show that $(AB)^2 = \frac{a^4}{(x_1)^2} + \frac{a^2 b^2}{a^2 - (x_1)^2}$
- (2)
- Given that $\frac{d^2}{d(x_1)^2} (AB)^2 > 0$ for $0 < |x| < a$, show that $(AB)^2$ is a minimum when $(x_1)^2 = \frac{a^3}{a+b}$
- (3)
- Hence, find the minimum length of AB
- (2)

(end of examination)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

ANSWERS Assess task 3 ext 2 2010

$$\begin{aligned}
 \textcircled{21} \quad (a) \quad & \int x (x^2+4)^{-\frac{1}{2}} dx \\
 &= +\frac{1}{2} \int 2x (x^2+4)^{-\frac{1}{2}} dx \quad (2) \\
 &= \frac{1}{2} \times (x^2+4)^{\frac{1}{2}} \times 2 + C \\
 &= \sqrt{x^2+4} + C
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int \frac{\cos^{-1} x}{\sqrt{1-x^2}} dx \\
 &= - \int \frac{-1}{\sqrt{1-x^2}} \cos^{-1} x dx \quad (2) \\
 &= -\frac{1}{2} (\cos^{-1} x)^2 + C
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \int \frac{1}{e^x + \frac{1}{e^x}} dx = \int \frac{1}{\frac{e^{2x}+1}{e^x}} dx \\
 &= \int \frac{e^x}{1+(e^x)^2} dx \quad (2) \\
 &= \tan^{-1}(e^x) + C
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & \int \frac{1}{\sqrt{x^2+4x-12}} dx \\
 &= \int \frac{1}{\sqrt{x^2+4x+4-16}} dx \\
 &= \int \frac{1}{\sqrt{(x+2)^2-16}} dx \\
 &\quad \text{let } u = x+2 \\
 &\quad \frac{du}{dx} = 1 \\
 &\quad du = dx \quad (3) \\
 &= \int \frac{1}{\sqrt{u^2-4^2}} du \\
 &= \ln \left| u + \sqrt{u^2-4^2} \right| + C \\
 &= \ln \left| x+2 + \sqrt{(x+2)^2-4^2} \right| + C
 \end{aligned}$$

$$(e) \int \frac{1}{5+3\cos x} dx$$

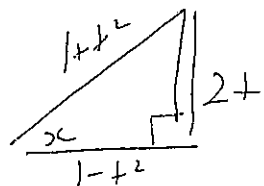
$$\text{Let } t = \tan \frac{x}{2}$$

$$t = \frac{1}{2} \cdot 2 \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \\ = \frac{1}{2} (1 + \tan^2 \frac{x}{2})$$

$$\frac{dt}{dx} = \frac{1+t^2}{2}$$

$$dx = \frac{2 dt}{1+t^2}$$



$$\therefore = \int \frac{1}{5+3\frac{(1-t^2)}{(1+t^2)}} \times \frac{2 dt}{1+t^2}$$

$$= \int \frac{1}{\frac{5+5+t^2+3-3t^2}{(1+t^2)}} \times \frac{2 dt}{(1+t^2)}$$

$$= \int \frac{(1+t^2)}{2(1+t^2)} \times \frac{2 dt}{(1+t^2)}$$

$$= \int \frac{1}{t^2+1} dt$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{t}{1} \right) + C$$

$$= \frac{1}{2} \tan^{-1} \left[\frac{\tan \frac{x}{2}}{1} \right] + C$$

(3)

$$(f) \text{ let } u = x^n$$

$$\frac{du}{dx} = nx^{n-1}$$

$$\frac{dv}{dx} = e^{-x}$$

$$v = -e^{-x}$$

$$= \left[x^n \cdot -e^{-x} \right]_0^2 + \int_0^2 e^{-x} \cdot nx^{n-1} dx$$

$$= -2^n e^{-2} + n \int_0^2 x^{n-1} e^{-x} dx$$

$$= -2^n e^{-2} + n I_{n-1}$$

(3)

$$I_2 = -2^2 e^{-2} + 2 I_1$$

$$= -4e^{-2} + 2[-2e^{-2} + I_0]$$

$$= -4e^{-2} - 4e^{-2} + 2 I_0$$

$$= -8e^{-2} + 2 \times \int_0^2 e^{-x} dx$$

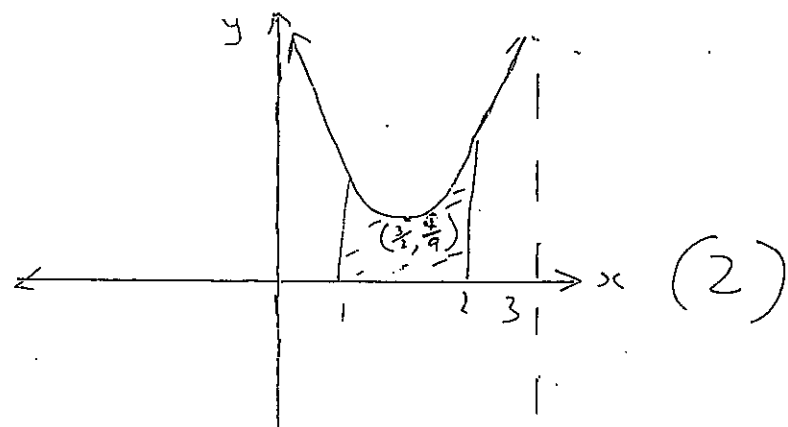
$$= -8e^{-2} + 2 \left[-e^{-x} \right]_0^2$$

$$= -8e^{-2} + 2(-e^{-2} - -1)$$

$$= -10e^{-2} + 2$$

(2)

(9)



(2)

It would be of some help to find any stat. pts.

Stat pts occur when $y' = 0$

$$y = (3x - x^2)^{-1}$$

$$y' = -(3x - x^2)^{-2} (3 - 2x)$$

$$= \frac{2x - 3}{(3x - x^2)^2}$$

$y' = 0$ when $x = 3/2$ $\therefore (3/2, 3/4)$ is a stat pt.

Test for nature:

x	1	$3/2$	2
y'	-ve	0	+ve

 $\therefore$ Min T.P.

$$\therefore \text{Area} = \int_0^2 \frac{1}{x(3-x)} dx$$

$$\frac{1}{x(3-x)} = \frac{a}{x} + \frac{b}{3-x}$$

$$= \frac{1}{3} \int_0^2 \left(\frac{1}{x} + \frac{1}{3-x} \right) dx$$

$$= \frac{1}{3} \left[\ln x - \ln |3-x| \right]_0^2 \quad \begin{matrix} (x=3) & b = \frac{1}{3} \\ (x=0) & a = \frac{1}{3} \end{matrix}$$

$$= \frac{1}{3} \left[(\ln 2 - 1) - (\ln 1 - \ln 2) \right] \therefore = \frac{1}{3x} + \frac{1}{3(3-x)}$$

$$= \frac{1}{3} (2 \ln 2) = \frac{2}{3} \ln 2 \text{ units}^2$$

Conics

Question 2

$$(a) \quad \frac{x^2}{4} + \frac{y^2}{3} = 1 \quad a = 2 \quad b = \sqrt{3}$$

$$(i) \quad b^2 = a^2(1 - e^2)$$

$$3 = 4(1 - e^2)$$

$$3 = 4 - 4e^2$$

$$4e^2 = 1$$

$$e^2 = \frac{1}{4}$$

$$e = \frac{1}{2}$$

(3)

$$(ii) \quad \text{Foci: } (\pm ae, 0) \\ = (\pm 1, 0)$$

$$(iii) \quad x = \pm \frac{a}{e} \\ = \pm 4$$

$$(b) \quad x^2 - 16y^2 = 16$$

$$\frac{x^2}{16} - y^2 = 1$$

$$a^2 = 16 \quad b^2 = 1 \\ (2, -4)$$

$$\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

$$\frac{2x}{16} + \frac{4y}{1} = 1$$

$$2x + 64y = 16$$

$$x + 32y = 8$$

(2)

$$(c) \quad x = a \cos \theta \quad y = b \sin \theta$$

$$\frac{x}{a} = \cos \theta \quad \frac{y}{b} = \sin \theta$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$$y^2 = b^2 \left(\frac{a^2 - x^2}{a^2}\right)$$

$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

$$\therefore A = 2 \int_{-a}^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= \frac{2b}{a} \left(\int_{-a}^a \sqrt{a^2 - x^2} dx \right)$$

Area of semi circle
so $\frac{2b}{a} \times \frac{\pi \times a^2}{2}$

$$= \frac{2\pi a^2 b}{2a}$$

$$= \pi ab$$

(4)

(d) Equation of normal is

$$px - \frac{y}{p} = c \left(p^2 - \frac{1}{p^2} \right)$$

$$\text{As } y = \frac{c^2}{x}$$

$$px - \frac{c^2}{px} = c \left(p^2 - \frac{1}{p^2} \right)$$

$$p^2 x^2 - c^2 = c p x \left(p^2 - \frac{1}{p^2} \right)$$

$$p^2 x^2 - c^2 = c p^3 x - \frac{c x}{p}$$

$$p^2 x^2 - c p^3 x + \frac{c x}{p} + c^2 = 0$$

$$p^2 x (x - c p) + \frac{c}{p} (x - c p) = 0$$

$$(x - cp)(p^2x + \frac{c}{p}) = 0$$

$$\therefore R(-\frac{c}{p^3}, -cp^3)$$

$$x = cp \text{ or } -\frac{c}{p^3}$$

$$m_{PQ} = \frac{\frac{c}{p} + \frac{c}{p}}{cp + cp}$$

$$= \frac{\frac{2c}{p}}{2cp}$$

$$= \frac{2c}{p} \times \frac{1}{2cp}$$

$$= \frac{1}{p^2}$$

$$m_{QR} = \frac{-\frac{c}{p} + cp^3}{-cp + \frac{c}{p^3}}$$

$$= \frac{-\frac{c + cp^4}{p}}{\frac{-cp^4 + c}{p^3}}$$

$$= \frac{c(p^4 - 1)}{\frac{-c(p^4 - 1)}{p^2}}$$

$$= -p^2$$

$$\therefore m_{PQ} \times m_{QR} = -1$$

$\therefore \angle PQR$ is a right angle.
(4)

(2) Equation of tangent is given by
 $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = 1$

cuts x axis when $y = 0$

$$\frac{x_1x}{a^2} = 1$$

$$x_1x = a^2$$

$$x = \frac{a^2}{x_1}$$

$$\therefore A(\frac{a^2}{x_1}, 0)$$

cuts y axis when $x = 0$

$$\therefore B(0, \frac{b^2}{y_1}) \quad (2)$$

$$(ii) (AB)^2 = \frac{a^4}{(x_1)^2} + \frac{b^4}{(y_1)^2}$$

$$\text{but } \frac{y_1^2}{b^2} = 1 - \frac{x_1^2}{a^2}$$

$$= \frac{a^2 - x_1^2}{a^2}$$

$$\text{so } \frac{b^2}{y_1^2} = \frac{a^2}{a^2 - x_1^2}$$

$$\therefore (AB)^2 = \frac{a^4}{(x_1)^2} + b^2 \left(\frac{a^2}{a^2 - x_1^2} \right) \quad (2)$$

$$(iii) \frac{d}{dx} (AB)^2 = \frac{-2a^4}{(x_1)^3} + \frac{a^2b^2(-1)(-2x_1)}{(a^2 - x_1^2)^2}$$

$$= \frac{-2a^4}{(x_1)^3} + \frac{2a^2b^2x_1}{(a^2 - x_1^2)^2}$$

$$= \frac{2a^2 [b^2 x_1^4 - a^2 (a^2 - x_1^2)^2]}{x_1^3 (a^2 - x_1^2)^2}$$

For st pts $\frac{d}{d(x_1)} (AB)^2 = 0$

$$\therefore b^2 x_1^4 - a^2 (a^2 - x_1^2)^2 = 0$$

$$[bx_1^2 - a(a^2 - x_1^2)][bx_1^2 + a(a^2 - x_1^2)] = 0$$

$$[(b+a)x_1^2 - a^3][(b-a)x_1^2 + a^3] = 0$$

$$\therefore x_1^2 = \frac{a^3}{(a+b)} \quad \text{OR} \quad x_1^2 = \frac{a^3}{(a-b)}$$

$$x_1^2 = a^2 \cdot \frac{a}{a-b} \quad (3)$$

which is greater than a^2

$\therefore$ outside domain

$$\therefore x_1^2 = \frac{a^3}{(a+b)} \Rightarrow a^2 \cdot \frac{a}{a+b} < a^2$$

$\therefore$ since $\frac{d^2}{d(x_1)^2} (AB)^2 > 0$ for $0 < |x| < a$

then $x_1^2 = \frac{a^3}{a+b}$ gives a minimum

$$(iv) (AB)^2 = \frac{a^4(a+b)}{a^3} + \frac{a^2 b^2}{a^2 - \frac{a^3}{a+b}}$$

$$= a(a+b) + \frac{a^2 b^2 (a+b)}{(a+b)a^2 - a^3}$$

$$= a(a+b) + \frac{a^2 b^2 (a+b)}{a^3 + a^2 b - a^3}$$

$$= a^2 + ab + b(a+b) \quad (2)$$

$$= a^2 + 2ab + b^2$$

$$= (a+b)^2$$

$\therefore$ Min. length is $a+b$